



SD oxide TFT design

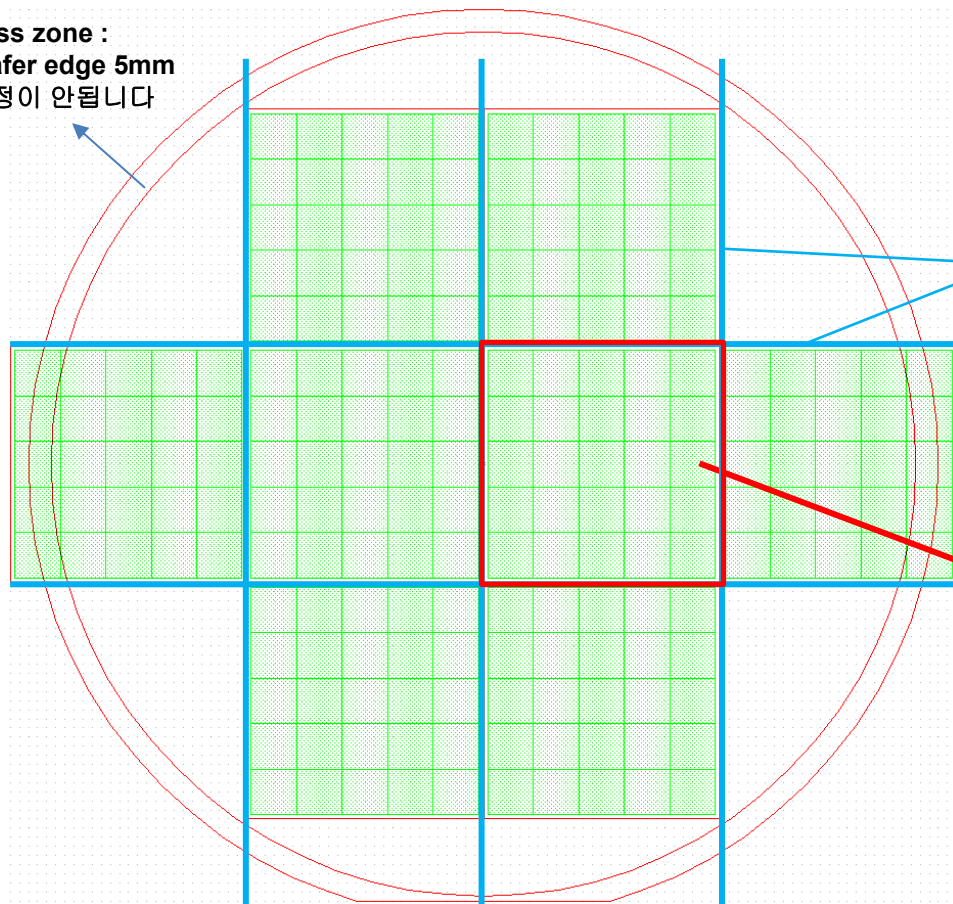
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Wafer and reticle

Loss zone :
Wafer edge 5mm
공정이 안됩니다

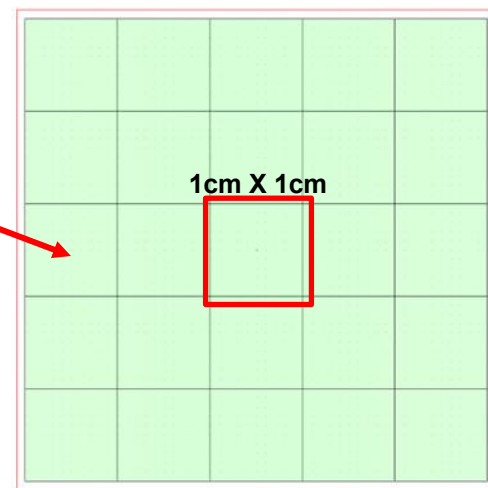


Wafer size : 8 inches (200mm diameter)

1 stepper reticle : 52000um x 52000um

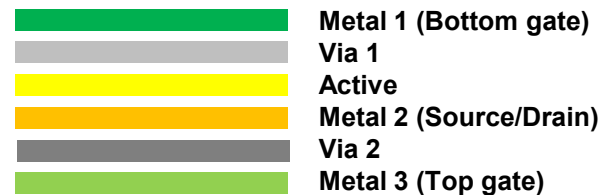
1 die에 1cm X 1cm cell이 5 x 5 로 들어간 경우

Wafer cut 시 100~150um 정도의 loss 발생



Layout diagram – 6um/4um TFT (split structure)

Metal – Layer # 커질 수록 더 넓게
Via – Layer # 커질 수록 더 좁게
Standard pad size: 200um x 200um
측정 장비의 스펙에 따라 변경 가능

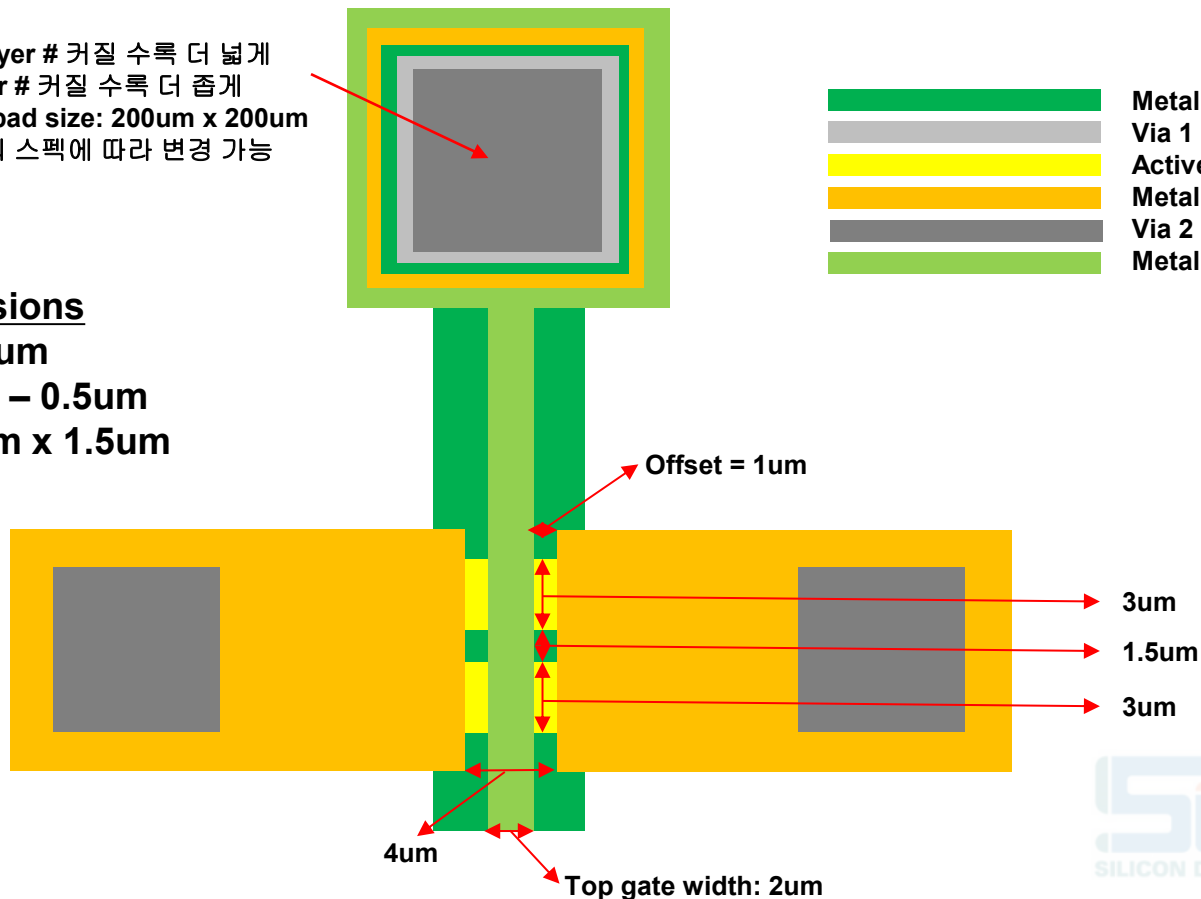


Critical dimensions

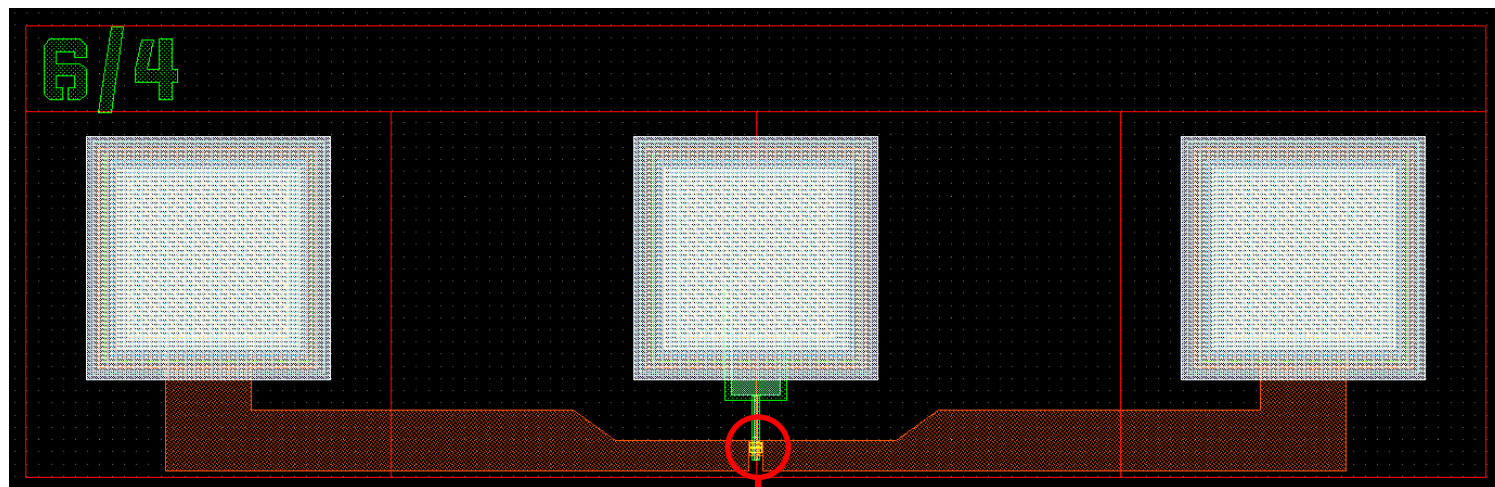
Metal lines – 1um

Offset spacing – 0.5um

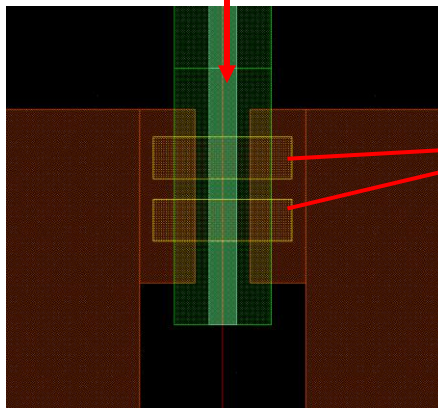
Via holes 1.5um x 1.5um



Layout – W = 6um, Top gate 2um, bottom gate 4um TFT



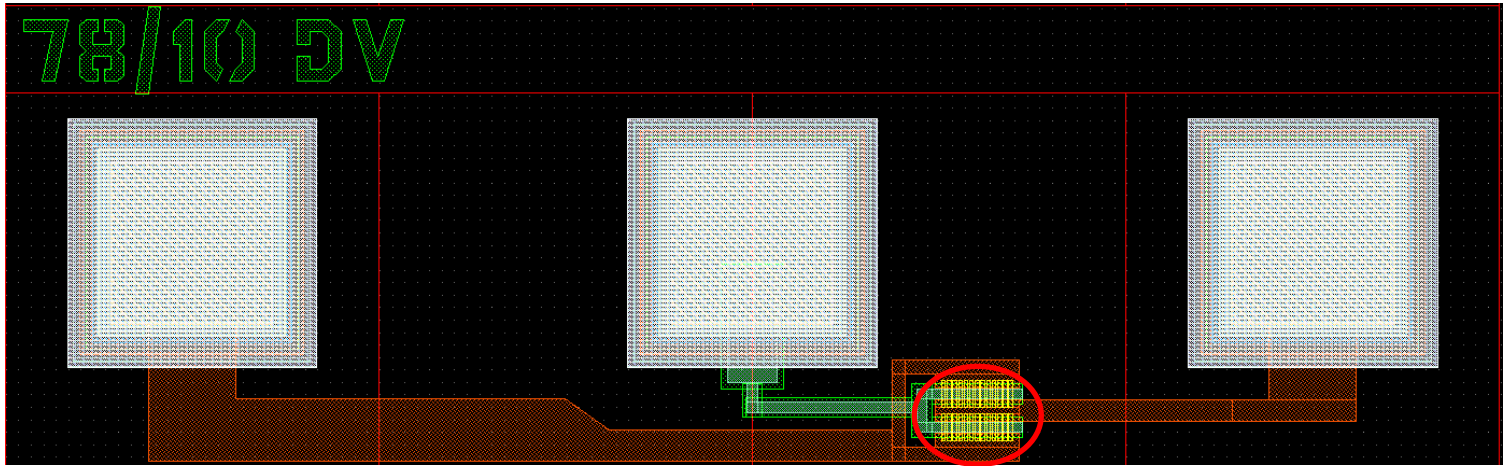
3um split structure
1um offset



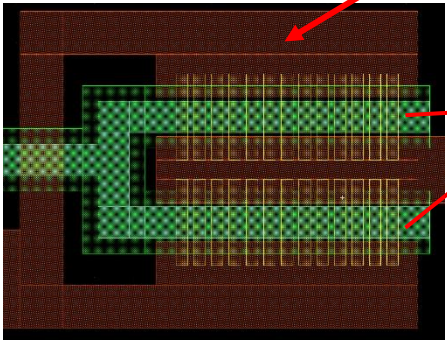
TFT Active 영역이 3um 2개로 split

SD
SILICON DISPLAY

Layout – W = 78um, Top gate 8um, bottom gate 10um TFT



3um split structure
1um offset



TFT Active 영역이 3um 영역으로 split



Process flow

Step #	Process
1	Buffer deposition
2	Bottom gate deposition
3	Bottom gate etch
4	Gate insulator deposition
5	Active deposition
6	Active etch
7	Via 1 etch
8	Data metal deposition
9	Data metal etch
10	Passivation 1
11	Via 2 etch
12	Top gate deposition
13	Top gate etch
14	Passivation 2
15	Via 3 etch
16	Pad metal deposition
17	Pad etch



Layer information

	Layer (GDS)	Material	Thickness (Å)	R_s	ϵ
#0	Dummy	-	-	-	-
#2	Bottom gate	Mo	120nm	$1.18\Omega/\square$	
#1	Active	Oxide TFT	20nm		
#3	GI (VIA1)	SiNx + SiO ₂	50nm + 75nm		$\epsilon = 7 + 3.9$
#4	Data	Mo	150nm	$0.66\Omega/\square$	
#5	Data passivation (VIA2)	SiO ₂	200nm (100 + 100)		$\epsilon = 3.9$
#6	Top gate	Mo	150nm	$0.66\Omega/\square$	
#7	Top gate passivation (VIA3)	SiO ₂	400nm		$\epsilon = 3.9$
#8	Pad (Optional)	Mo	400nm	$0.32\Omega/\square$	



Cross section 2: Bottom gate deposition

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)

Bottom gate

Substrate

Cross section 3: Bottom gate etch

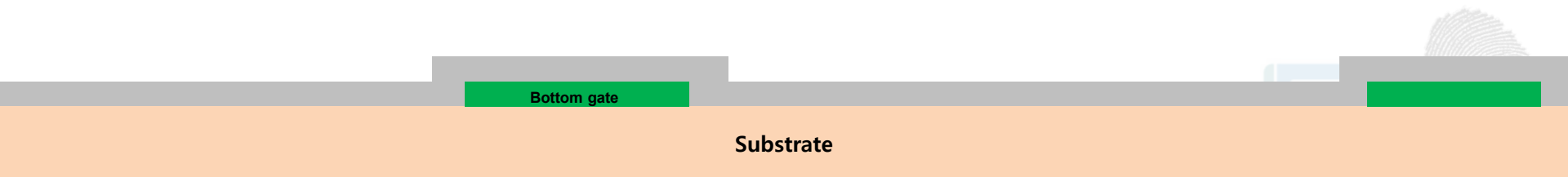
	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)

Bottom gate

Substrate

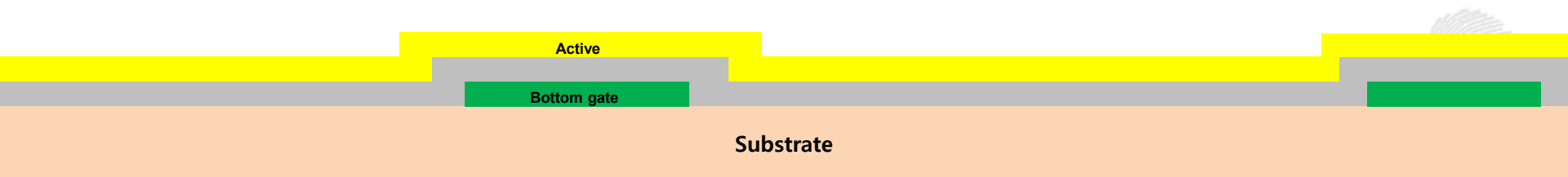
Cross section 4: Gate insulator deposition

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



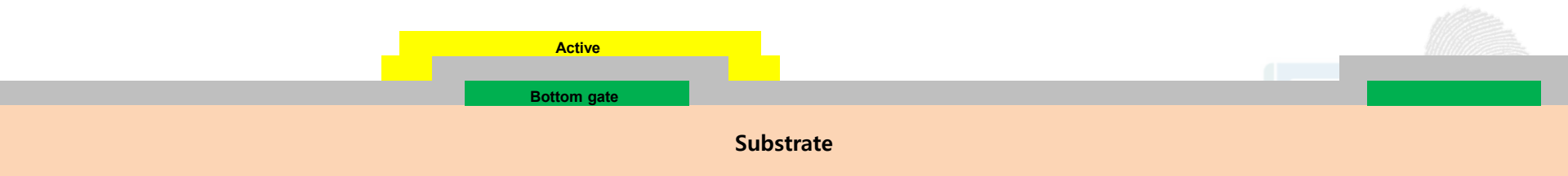
Cross section 5: Active deposition

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



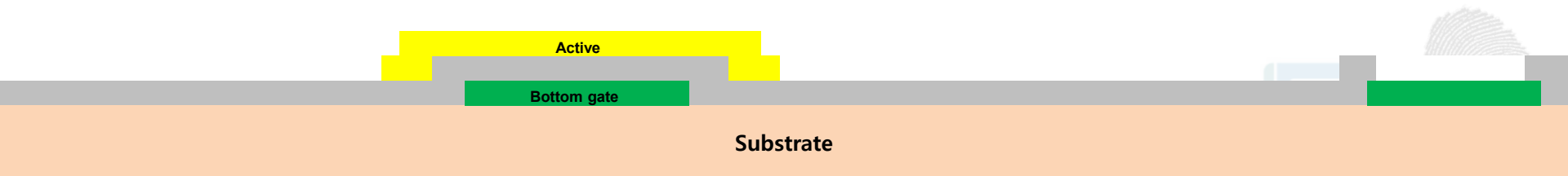
Cross section 6: Active etch

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



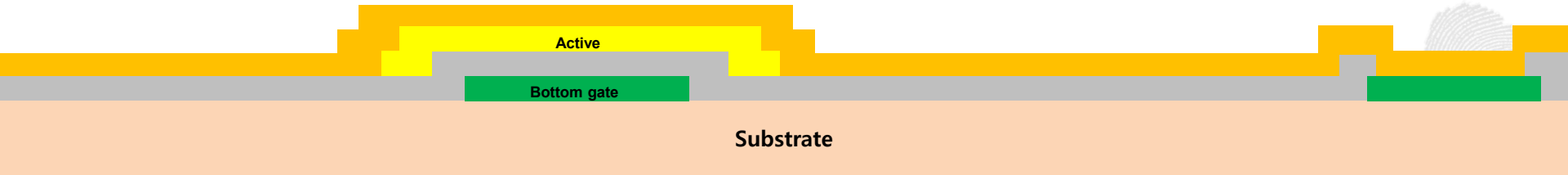
Cross section 7: Via 1 etch

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



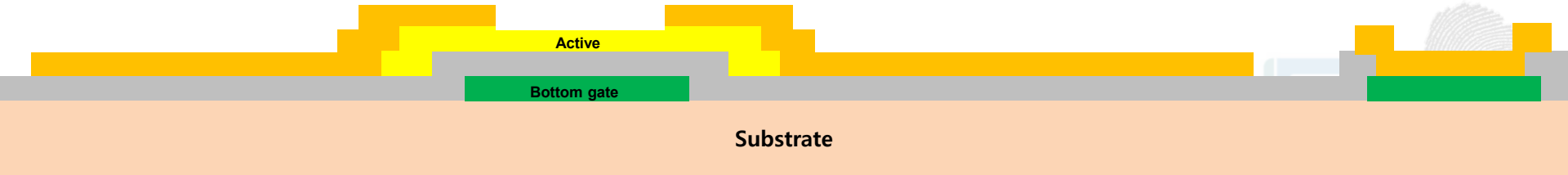
Cross section 8: Metal 2 deposition (data)

- Metal 1 (Bottom gate)
- Gate insulator
- Active
- Metal 2 (Source/Drain)
- Passivation 1
- Metal 3 (Top gate)
- Passivation 2
- Pad (optional)



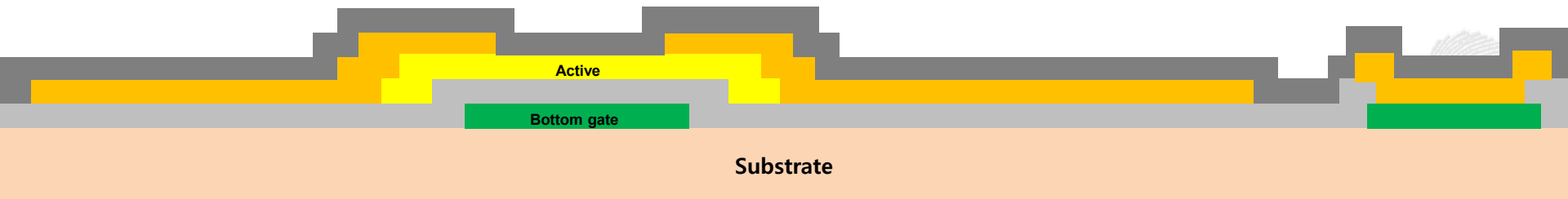
Cross section 9: Metal 2 etch

- Metal 1 (Bottom gate)
- Gate insulator
- Active
- Metal 2 (Source/Drain)
- Passivation 1
- Metal 3 (Top gate)
- Passivation 2
- Pad (optional)



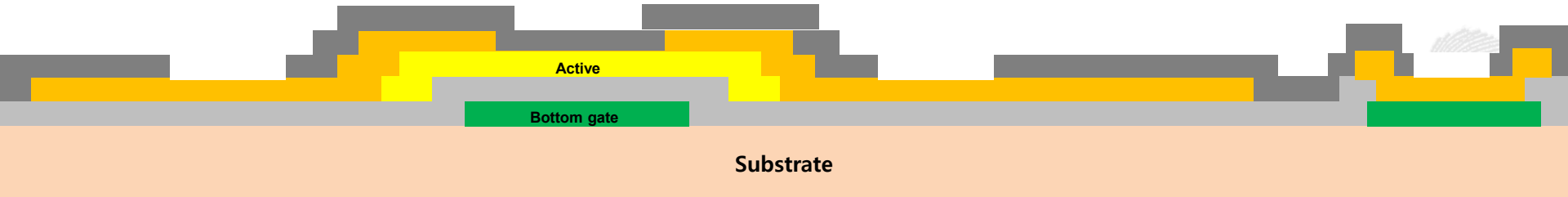
Cross section 10: Passivation 1 deposition

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



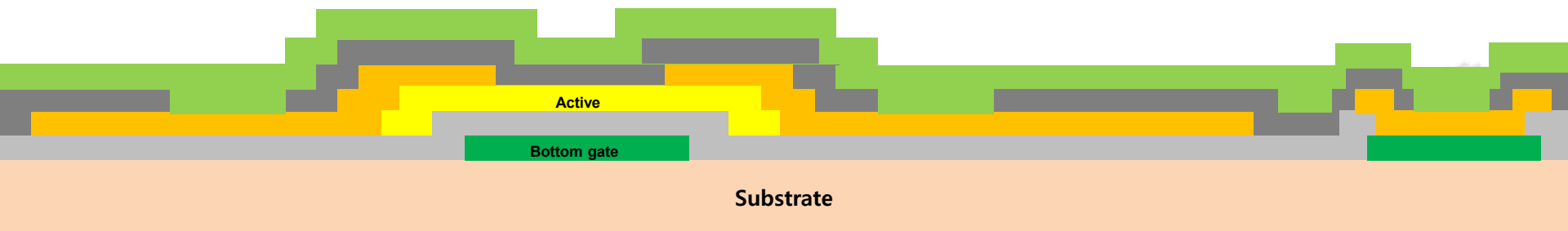
Cross section 11: Via 2etch

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



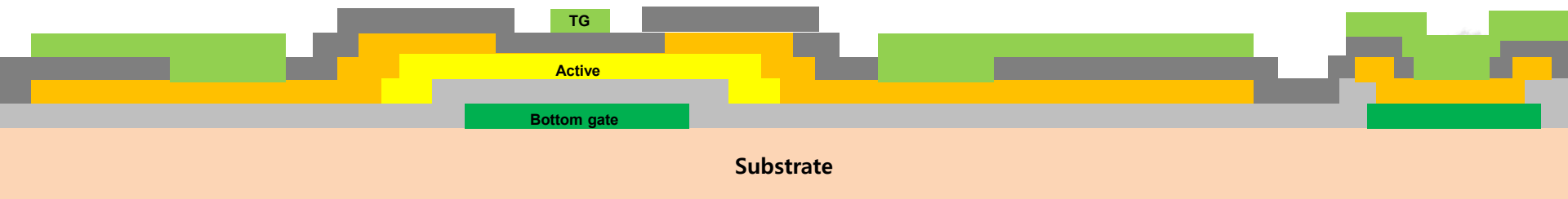
Cross section 12: Top gate deposition

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



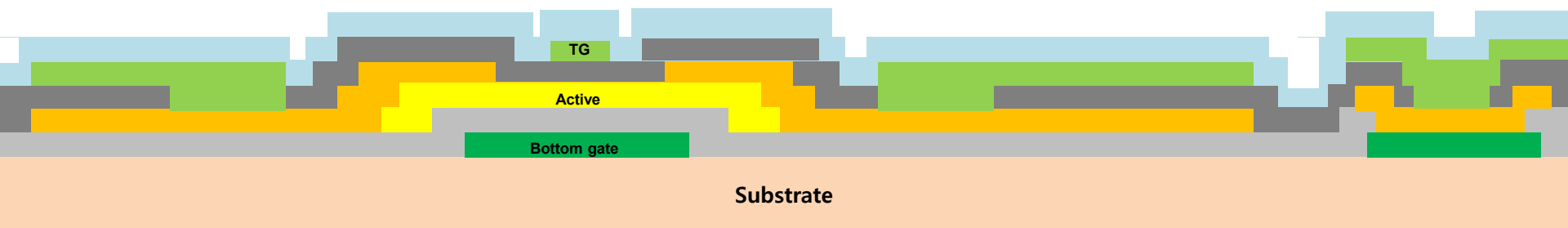
Cross section 13: Top gate etch

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



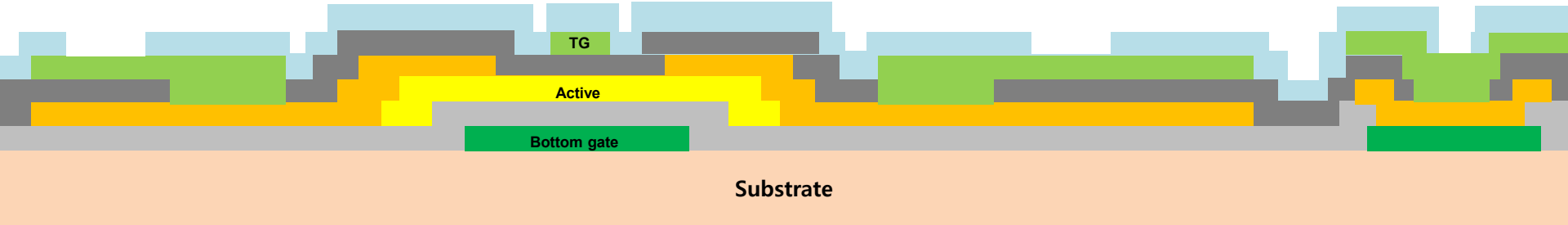
Cross section 14: Passivation 2 deposition

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



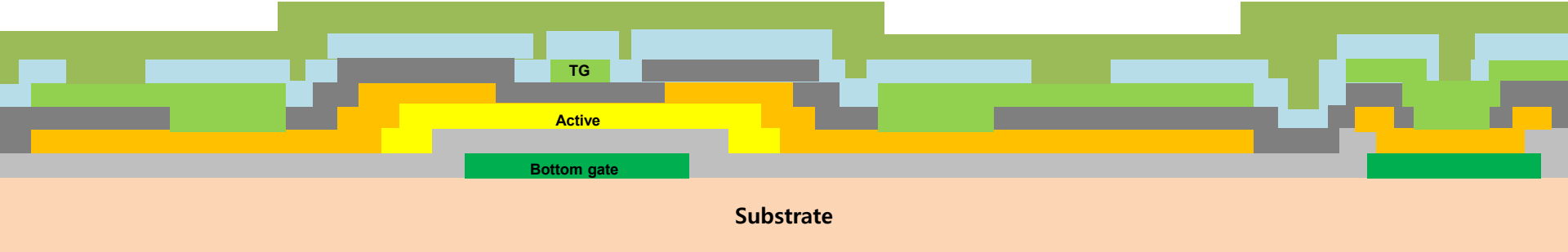
Cross section 14: Via 3 etch

- Metal 1 (Bottom gate)
- Gate insulator
- Active
- Metal 2 (Source/Drain)
- Passivation 1
- Metal 3 (Top gate)
- Passivation 2
- Pad (optional)



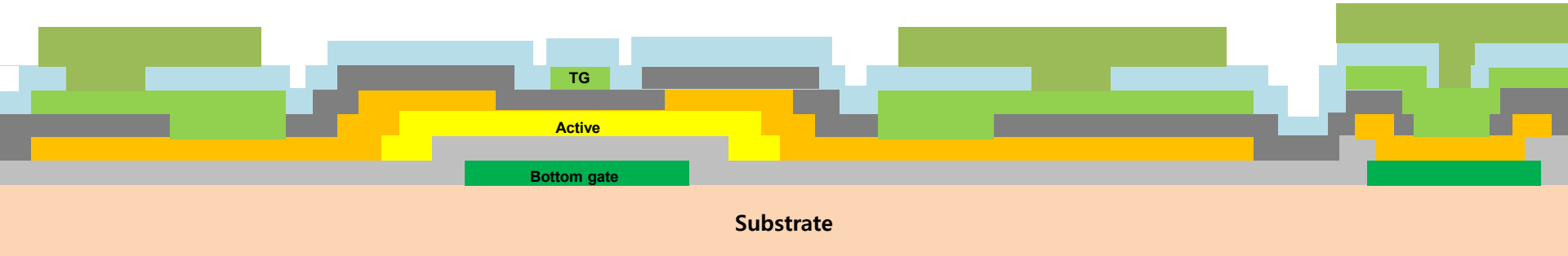
Cross section 14: Pad 3 deposition

- Metal 1 (Bottom gate)
- Gate insulator
- Active
- Metal 2 (Source/Drain)
- Passivation 1
- Metal 3 (Top gate)
- Passivation 2
- Pad (optional)



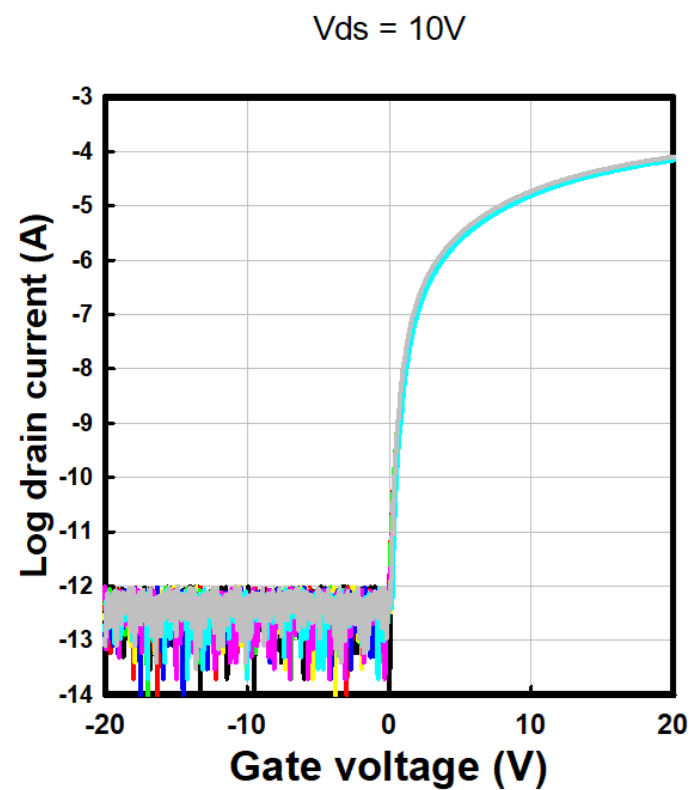
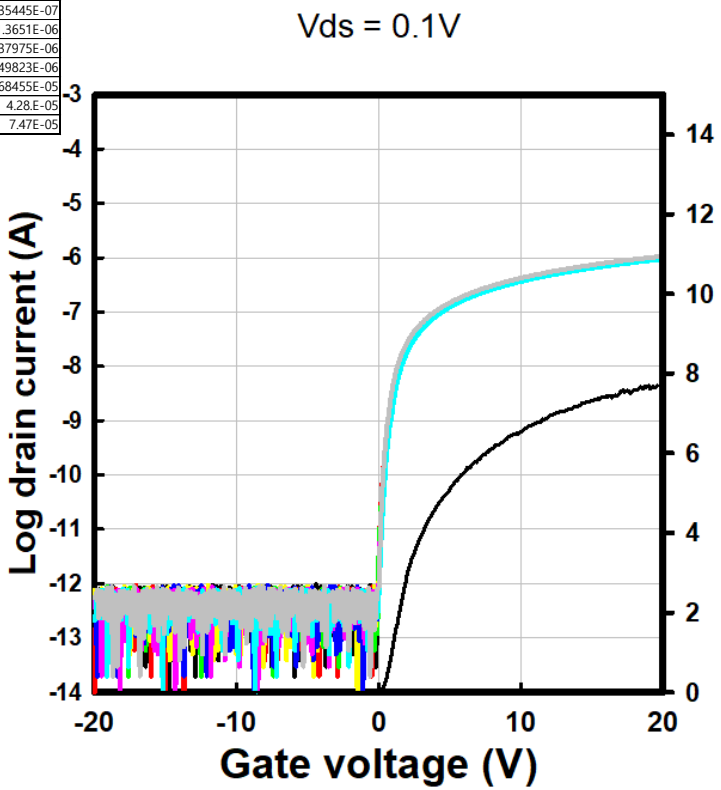
Cross section 14: Pad etch

	Metal 1 (Bottom gate)
	Gate insulator
	Active
	Metal 2 (Source/Drain)
	Passivation 1
	Metal 3 (Top gate)
	Passivation 2
	Pad (optional)



Oxide BCE TFT W=6um, bottom gate 4um, top gate 2um offset:1um Transfe

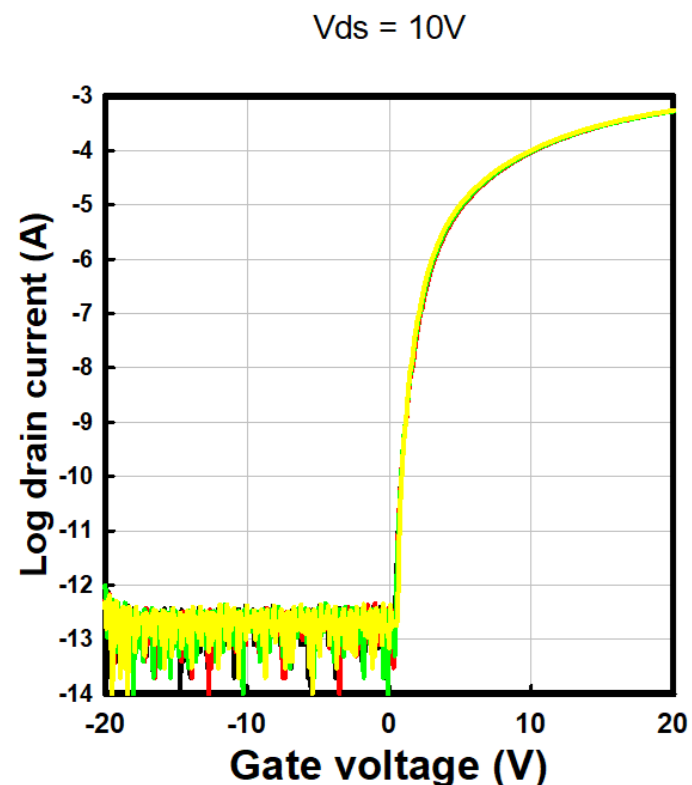
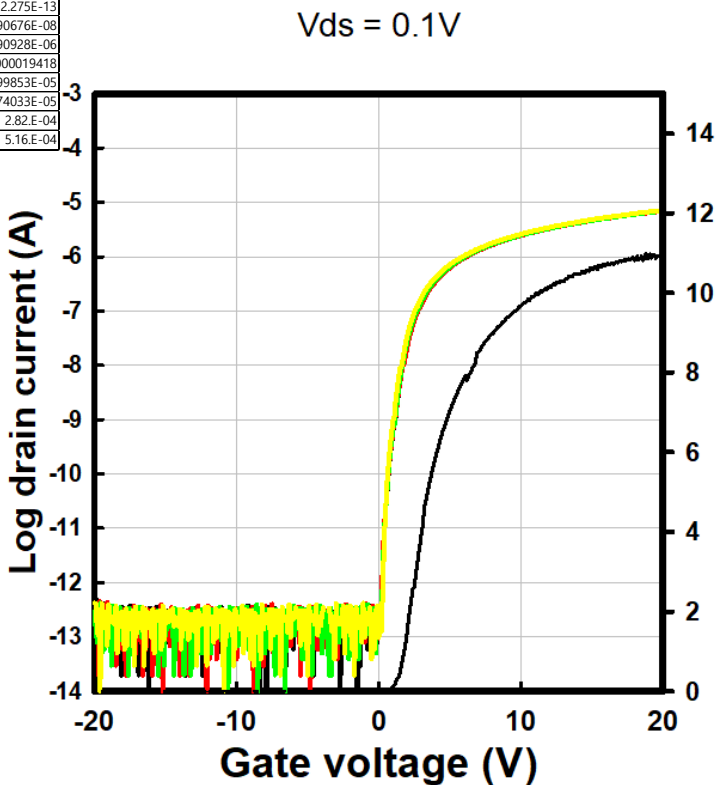
V_gate	Vds = 0.1V	Vds = 10V
-20	1.00E-13	1.00E-13
-10	1.00E-13	1.00E-13
0	1.9575E-12	4.475E-13
2	2.1382E-08	1.35445E-07
4	8.90645E-08	1.3651E-06
6	1.7693E-07	4.37975E-06
8	2.77393E-07	9.49823E-06
10	3.86338E-07	1.68455E-05
15	6.84003E-07	4.28E-05
20	1.00422E-06	7.47E-05



Vth = 1.55V, SS = 0.270

Oxide BCE TFT W=78um, bottom gate 10um, top gate 8um offset:1um Transfer

V_gate	Vds = 0.1V	Vds = 10V
-20	1.00E-13	1.00E-13
-10	1.00E-13	1.00E-13
0	2.125E-13	2.275E-13
2	5.17658E-08	6.90676E-08
4	4.51333E-07	3.90928E-06
6	1.04462E-06	0.000019418
8	1.73928E-06	4.99853E-05
10	2.5053E-06	9.74033E-05
15	4.60395E-06	2.82 E-04
20	6.85578E-06	5.16 E-04



Vth = 1.7V, SS = 0.221

Simulation parameters

N type TFT model parameters (6/4 TFT) SILVACO SmartSPICE

.MODEL igzo ntft (
+LEVEL = 35
+TOX = 2.50E-7 EPSI = 4.74 TNOM = 27
+VTO = 1.25 VMIN = 0.45 DELTA = 2.2
+ALPHASAT= 0.455 M = 2.1 LAMBDA = 15E-3
+MUBAND = 0.10 VAA = 7200 GAMMA = 0.788
+VFB = -1.8 EPS = 3.71 VO = 0.155
+GMIN = 1.02E23 IOL = 3E-15
+VDSL = 100 VGSL = 10 SIGMAO =7E-15
+CGSO = 2.518E-10 CGDO = 2.518E-10)

W = 6um
TG = 2um
BG = 4um TFT

V_gate	Vds = 0.1V	Vds = 10V
-20	1.00E-13	1.00E-13
-10	1.00E-13	1.00E-13
0	1.9575E-12	4.475E-13
2	2.1382E-08	1.35445E-07
4	8.90645E-08	1.3651E-06
6	1.7693E-07	4.37975E-06
8	2.77393E-07	9.49823E-06
10	3.86338E-07	1.68455E-05
15	6.84003E-07	4.28.E-05
20	1.00422E-06	7.47E-05

W = 78um
TG = 8um
BG = 10um TFT

V_gate	Vds = 0.1V	Vds = 10V
-20	1.00E-13	1.00E-13
-10	1.00E-13	1.00E-13
0	2.125E-13	2.275E-13
2	5.17658E-08	6.90676E-08
4	4.51333E-07	3.90928E-06
6	1.04462E-06	0.000019418
8	1.73928E-06	4.99853E-05
10	2.5053E-06	9.74033E-05
15	4.60395E-06	2.82.E-04
20	6.85578E-06	5.16.E-04